

Programming for Computational Linguistics 2025/2026

Exercise Set 9 (2026-01-08) Code Styles

Once you have completed all exercises, send an email to the `progclgrader@ims.uni-stuttgart.de` with the title “submit set9”, and with your python files as attachments. You should receive a response with feedback and a grade.

Exercises are due on **2026-01-27 23:59**, but we encourage you to do them during the lab session.

Exercise 1. Read the article: <https://realpython.com/python-pep8/>. Do not submit anything, there are no points for this exercise.

Exercise 2. The file `happiness_2018.txt` contains a list of countries and their happiness scores in 2018 (see https://en.wikipedia.org/wiki/World_Happiness_Report):

```
1,Afghanistan,3.632,Asia
2,Albania,4.586,Europe
3,Algeria,5.295,Africa
4,Angola,3.795,Africa
5,Argentina,6.388,Latin America and Caribbeans
6,Armenia,4.321,Europe
7,Australia,7.272,Oceania
8,Austria,7.139,Europe
9,Azerbaijan,5.201,Europe
(...)
```

In a file `happy.py` write a function `find_happiest(filename)` which will read the given file with happiness scores and return a dictionary with the most happy country per continent. For example, for the file `happiness_2018.txt` your function should return:

```
print(find_happiest("happiness_2018.txt"))
```

```
{'Asia': 'Israel',
 'Europe': 'Finland',
 'Africa': 'Mauritius',
 'Latin America and Caribbeans': 'Costa Rica',
 'Oceania': 'New Zealand',
 'North America': 'Canada'
}
```

The grader will check your function on reports from other years, so do not hard-code the solution for 2018. **Be sure that your code follows recommendations of PEP-8** – the grader will use `pycodestyle` to check your code and it will not give you points if there will be any errors or warnings.

Exercise 3. https://en.wikipedia.org/wiki/Substitution_cipher:

In cryptography, a substitution cipher is a method of encrypting by which units of plaintext are replaced with ciphertext, according to a fixed system; the "units" may be single letters (the most common), pairs of letters, triplets of letters, mixtures of the above, and so forth. The receiver deciphers the text by performing the inverse substitution.

In this exercise we will consider a very easy cipher method. The method will substitute letters of an alphabet `abcdefghijklmnopqrstuvwxyz` with letters on the corresponding positions of a given secret permutation. For example, for the secret permutation `wnoegbjpkyxlfiuastqhvmcrzd` the cipher method will replace letters accordingly:

a	→	w	n	→	i
b	→	n	o	→	u
c	→	o	p	→	a
d	→	e	q	→	s
e	→	g	r	→	t
f	→	b	s	→	q
g	→	j	t	→	h
h	→	p	u	→	v
i	→	k	v	→	m
j	→	y	w	→	c
k	→	x	x	→	r
l	→	l	y	→	z
m	→	f	z	→	d

So an enciphered version of message "i love python" will be "k lung azhpui". In a file `cipher.py` write a function `decipher(msg, perm)` which will decipher the given message `msg` according to the given `perm`. **Be sure that your code follows recommendations of PEP-8.**

Example usage of your function:

```
perm = "wnoegbjpkylfiuastqhvmcrzd"
print(decipher("wnoeg", perm))
print(decipher("rzd", perm))
print(decipher("azhpui", perm))
print(decipher("hpg ntuci bur yvfaq umgt hpg euj", perm))
```

```
abcde
xyz
python
the brown fox jumps over the dog
```

You can assume that all the letters of the message are lowercased, but the message can contain spaces.

Exercise 4. This exercise is worth 20 points – an equivalent of two regular exercises. As this exercise will be graded manually, you will receive no autograder feedback. We will only grade your final submission, and you will receive your grade after the submission deadline.

Of course your journey with Python should not end with the end of this class. We want to encourage you to find a way to improve your skills. One way might be... becoming a coding warrior :>

Register on <https://www.codewars.com> and reach the level of **7 kyu**. Submit a screenshot of your profile with a visible **profile id** and information about your progress (the profile id should be something that links directly to you, e.g., your name, nickname from the grader). Name the picture **warrior.png**. Such a picture might look like this:

